Project Report

On

Design and MATLAB Simulation of a Passive Two-Port Filter

Course No: ECE 2108

Course Name: Signal and Systems Laboratory

Submitted by:

Kankon Mondal	2409021
Mahsina Khan Nabila	2409024

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Department of Electronics and Communication Engineering

Khulna University of Engineering & Technology

Khulna-9203, Bangladesh.

Problem Statement

Design and simulate a passive two-port RC filter network to preserve a useful low-frequency sensor signal while attenuating unwanted high-frequency interference. The project should demonstrate the application of two-port network analysis, circuit design principles, and MATLAB simulation to evaluate and optimize the filter's performance under different operating conditions.

Open-ended features:

1. Use Propose two different passive RC filter arrangements and derive their ABCD parameters, along with either Z or Y parameters for one design.
2. Use MATLAB to simulate and plot the magnitude and phase responses over a frequency range of 100 Hz to 100 kHz.
3. Select suitable resistor and capacitor values to achieve a gain of at least 0.80 at 1 kHz and at most 0.25 at 10 kHz.
4. Compare the proposed designs in terms of useful-signal preservation, interference rejection, and component count.
5. Investigate the effects of different load resistances (5 kΩ, 10 kΩ, and 20 kΩ) and ±5% variations in one resistor and one capacitor.

Result analysis:

1. Analyze the performance of the proposed filter designs at the useful-signal and interference frequencies.
2. Compare the simulated frequency responses and identify the component combinations that satisfy the specified gain requirements.
3. Discuss the results of the independent calculation and analyze the effect of the design revision.
4. Explain the limitations of the selected design and suggest an improvement if the interference frequency changes from 10 kHz to 5 kHz.

Evaluation

	Excellent	Above Average	Average	Below Average	Marks (10)
Organization/Presentation					
Necessary data presentation					
Result Analysis					
Conclusion					

Signature of Examiner

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1. Objectives

1. To design two passive RC low-pass filter circuits that satisfy the specified gain requirements at 1 kHz and 10 kHz.
2. To derive the ABCD parameters of both filter networks and the Z-parameters of at least one design.
3. To simulate and analyze the magnitude and phase responses over the frequency range of 100 Hz to 100 kHz using MATLAB.
4. To compare candidate component values and evaluate the effects of load resistance and ±5% component variations on filter performance.
5. To verify simulation results through theoretical calculations and revise a design to improve or evaluate its performance.

2. Introduction

Passive filters are essential electronic circuits used to select desired frequency components and suppress unwanted signals. They are widely applied in communication systems, measurement instruments, and signal-processing circuits. Among different filter configurations, the RC low-pass filter is a simple and economical solution that uses resistors and capacitors to attenuate higher-frequency signals while allowing lower-frequency signals to pass with comparatively less attenuation.

This project focuses on the design and MATLAB simulation of two passive RC low-pass filter networks with different configurations. The first design consists of a single-stage RC network, while the second employs two cascaded RC stages to achieve greater attenuation at higher frequencies. Both designs are evaluated considering a source resistance of $100\ \Omega$ and a nominal load resistance of $10\ k\Omega$.

The primary objective is to satisfy the specified gain requirements: a voltage gain magnitude of at least 0.80 at the useful frequency of 1 kHz and no more than 0.25 at the interference frequency of 10 kHz. MATLAB is used to analyze the magnitude and phase responses over the frequency range of 100 Hz to 100 kHz and to compare different component combinations. The analysis also investigates the effects of load resistance and ±5% variations in resistor and capacitor values on filter performance.

Furthermore, the project includes the derivation of ABCD parameters for both two-port networks and Z-parameters for at least one design. The theoretical analysis is compared with simulation results through hand calculations at selected frequencies. Finally, the two designs are compared in terms of frequency selectivity, gain requirements, and sensitivity to external and component variations to identify their respective advantages and limitations.

3. Theory

The theoretical analysis of passive RC filters is essential for understanding their frequency-dependent behavior and predicting their performance before simulation. This section presents the fundamental principles of RC low-pass filters, the mathematical derivation of the transfer functions for both proposed designs, and the two-port network parameters required for circuit analysis. These theoretical relationships provide the foundation for calculating voltage gain and phase response, selecting suitable component values, and verifying the simulation results against the specified design requirements.

3.1 Principle of Passive RC Low-Pass Filters

A passive RC low-pass filter consists of resistors and capacitors arranged to pass low-frequency signals while attenuating high-frequency components. The operation depends on the frequency-dependent impedance of the capacitor, given by

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j2\pi f C}$$

where f is the signal frequency, C is the capacitance, and $\omega = 2\pi f$ is the angular frequency. At low frequencies, the capacitor has a relatively high impedance, whereas at high frequencies, its impedance decreases, causing more of the signal to be attenuated.

The voltage gain of the filter is defined as

$$H(j\omega) = \frac{V_o}{V_s}$$

where V_s is the source voltage and V_o is the output voltage across the load. The gain magnitude and phase are expressed as

$$|H(j\omega)| = \left|\frac{V_o}{V_s}\right|$$
$$\phi(\omega) = \arg[H(j\omega)]$$

For this project, the design requirements are

$$|H(j2\pi \times 1000)| \geq 0.80$$
$$|H(j2\pi \times 10000)| \leq 0.25$$

These conditions ensure adequate transmission at the useful frequency and sufficient attenuation at the interference frequency.

3.2 Single-Stage RC Low-Pass Filter

Single-Stage consists of a series resistor R, a shunt capacitor C, source resistance R_s, and load resistance R_L. Applying Kirchhoff's laws and the voltage-divider principle, the transfer function is obtained as

$$H_A(s) = \frac{V_o(s)}{V_s(s)} = \frac{R_L}{R_s + R + R_L + sC(R_s + R)R_L}$$

For sinusoidal input signals, $s = j\omega$. Therefore, the magnitude of the transfer function is

$$|H_A(j\omega)| = \frac{R_L}{\sqrt{(R_s + R + R_L)^2 + [\omega C(R_s + R)R_L]^2}}$$

The phase response is

$$\phi_A(\omega) = -\tan^{-1}\left(\frac{\omega C(R_s + R)R_L}{R_s + R + R_L}\right)$$

The effective cutoff frequency, considering the source and load resistances, is

$$f_c = \frac{1}{2\pi C[(R_s + R) \parallel R_L]}$$

where $\parallel$ denotes the parallel combination of two resistances. Increasing Ror C generally increases attenuation at higher frequencies, although the source and load resistances also affect the overall gain.

3.3 Dual-Stage RC Low-Pass Filter

Dual-Stage consists of two cascaded RC sections, with series resistors $R_1$and R_2, and shunt capacitors C_1 and C_2. Since the stages are directly connected, the loading effect between stages must be considered rather than treating the sections as independent filters.

Using the ABCD transmission-parameter representation, the individual series-resistor and shunt-capacitor matrices are

$$T_R = \begin{bmatrix} 1 & R \\ 0 & 1 \end{bmatrix}, \qquad T_C = \begin{bmatrix} 1 & 0 \\ sC & 1 \end{bmatrix}$$

The overall transmission matrix is obtained by multiplying the matrices in cascade order:

$$T_B = T_{R_1} T_{C_1} T_{R_2} T_{C_2} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

The resulting parameters are

$$A = 1 + s(R_1 C_1 + R_1 C_2 + R_2 C_2) + s^2 R_1 R_2 C_1 C_2$$

$$B = R_1 + R_2 + s R_1 R_2 C_1$$

$$C = s(C_1 + C_2) + s^2 R_2 C_1 C_2$$

$$D = 1 + s R_2 C_1$$

With the load connected to port 2, the voltage transfer function, including the source resistance, is

$$H_B(s) = \frac{V_o}{V_s} = \frac{R_L}{A R_L + B + R_s(C R_L + D)}$$

The two-stage configuration can provide stronger high-frequency attenuation than a single-stage design. However, its performance depends on the component values, stage interaction, and load resistance.

3.4 Two-Port Parameters and Design Verification

For a linear two-port network, the ABCD parameters relate the input and output variables according to

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix}$$

Here,I_1enters port 1 and I_2 leaves port 2 toward the load. For Single-Stage, the Z-parameter representation, with both port currents defined as entering the network, is

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

where

$$Z_{11} = R + \frac{1}{sC}, \qquad Z_{12} = Z_{21} = Z_{22} = \frac{1}{sC}$$

Theoretical performance is verified by substituting $s = j2\pi f$ into the transfer functions and calculating the gain magnitude at the specified frequencies. The calculated results are then compared with MATLAB or Multisim results. Additional verification is performed by varying the

load resistance and changing resistor and capacitor values by $\pm 5\%$, allowing the stability and practical limitations of both designs to be assessed.

In summary, the theoretical equations developed in this section establish the basis for evaluating both filter designs under the given operating conditions. The transfer functions allow the voltage gain and phase response to be determined at different frequencies, while the two-port parameters provide a systematic method for analyzing the networks. These equations will be used in the subsequent MATLAB simulations, component-selection comparisons, load-variation analysis, and tolerance studies to assess whether the proposed designs satisfy the project specifications.

4. Experimental Setup

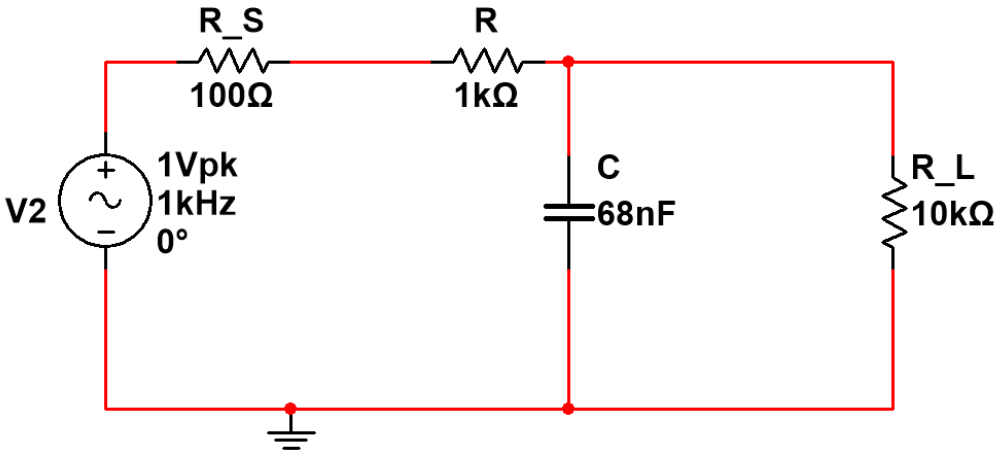


Fig-01: Circuit Diagram for Single-Stage RC Low-Pass Filter

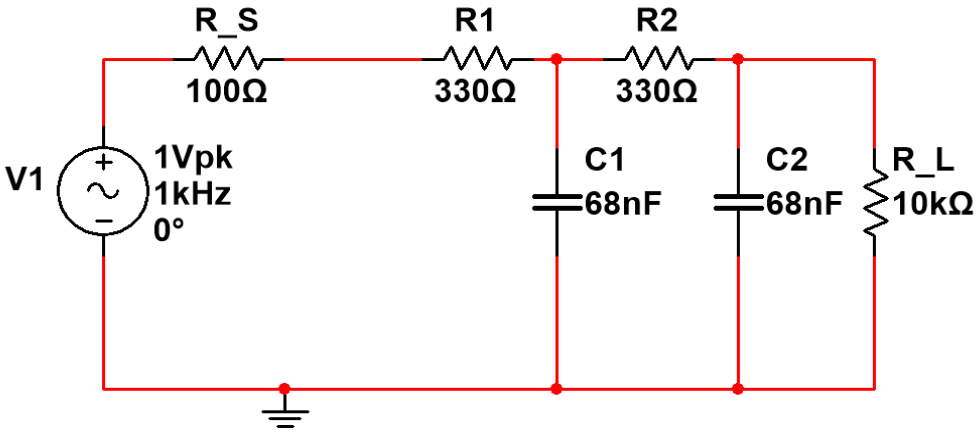


Fig-02: Circuit Diagram for Dual-Stage RC Low-Pass Filter

5. Experimental Data

Table-01: Performance Comparison of Tested RC Component Combinations for Single-Stage

R	C	V_s	V_{out} at 1 kHz	V_{out} at 10 kHz	Gain at 1 kHz	Gain at 10 kHz
470 Ω	33 nF	1	0.94	0.63	0.94	0.63
470 Ω	47 nF	1	0.934	0.504	0.934	0.504
470 Ω	68 nF	1	0.92	0.38	0.92	0.38
680 Ω	33 nF	1	0.92	0.51	0.92	0.51
680 Ω	47 nF	1	0.91	0.39	0.91	0.39
680 Ω	68 nF	1	0.89	0.285	0.89	0.285
1 kΩ	33 nF	1	0.88	0.394	0.88	0.394
1 kΩ	47 nF	1	0.864	0.291	0.864	0.291
1 kΩ	68 nF	1	0.83	0.207	0.83	0.207
1.5 kΩ	47 nF	1	0.8	0.205	0.8	0.205

Table-02: Performance Comparison of Tested RC Component Combinations for Dual-Stage

$R_1 = R_2$	$C_1 = C_2$	V_s	V_{out} at 1 kHz	V_{out} at 10 kHz	Gain at 1 kHz	Gain at 10 kHz
330 Ω	56 nF	1	0.877	0.233	0.877	0.233
330 Ω	68 nF	1	0.855	0.186	0.855	0.186
390 Ω	47 nF	1	0.872	0.241	0.872	0.241
390 Ω	56 nF	1	0.854	0.198	0.854	0.198
390 Ω	68 nF	1	0.827	0.156	0.827	0.156
470 Ω	47 nF	1	0.845	0.2	0.845	0.2
470 Ω	56 nF	1	0.822	0.162	0.822	0.162
470 Ω	68 nF	1	0.792	0.13	0.792	0.13
560 Ω	47 nF	1	0.814	0.166	0.814	0.166

Table-03: Selected Designs for the Two-Port RC Filter

	Single-Stage	***Dual-Stage***
Component values	1 kΩ, 68 nF	330 Ω , 68 nF
Load resistance	10 kΩ	10 kΩ
Gain at 1 kHz	0.830	0.855
Gain at 10 kHz	0.207	0.186
1 kHz requirement	Pass	Pass
10 kHz requirement	Pass	Pass

Table-04: Effect of Load Resistance on the RC Filters

	Single Stage		Double Stage	
Load Verification R_L	**1kHz**	**10kHz**	**1kHz**	**10kHz**
10k	0.83	0.207	0.855	0.186
5k	0.765	0.206	0.806	0.185
20k	0.865	0.207	0.885	0.2

Table-05: Effect of Component Tolerances on the RC Filters

	Single Stage		Double Stage	
5% Analysis	**1kHz**	**10kHz**	**1kHz**	**10kHz**
-5% R	0.838	0.216	0.863	0.196
+5% R	0.82	0.198	0.847	0.177
-5% C	0.835	0.217	0.854	0.188
+5% C	0.823	0.198	0.84	0.167

6. Calculations

6.1 Single-Stage RC Low-Pass Filter

6.1.1 Given parameters

$$\mathrm{R_s} = 100\,\Omega$$
$$\mathrm{R} = 1000\,\Omega$$
$$\mathrm{C} = 68 \times 10^{-9}\,\mathrm{F}$$
$$\mathrm{R_L} = 10000\,\Omega$$

The two specified frequencies are

$$\mathrm{f_1} = 1\,\mathrm{kHz}, \qquad \mathrm{f_2} = 10\,\mathrm{kHz}$$

Using $\omega = 2\pi f$,

$$\omega_1 = 6283.19\,\mathrm{rad/s}$$
$$\omega_2 = 62831.85\,\mathrm{rad/s}$$

6.1.2 Derivation of the transfer function

The impedance of the capacitor in the Laplace domain is

$$Z_C = \frac{1}{sC}$$

Since the capacitor is connected in parallel with the load resistor,

$$Z_p = Z_C \parallel R_L = \frac{R_L}{1 + sCR_L}$$

Applying the voltage-divider principle,

$$H_A(s) = \frac{V_o(s)}{V_s(s)} = \frac{Z_p}{R_s + R + Z_p}$$

Substituting Z_p,

$$H_A(s) = \frac{\frac{R_L}{1 + sCR_L}}{R_s + R + \frac{R_L}{1 + sCR_L}}$$

Simplifying,

$$H_A(s) = \frac{R_L}{R_s + R + R_L + sC(R_s + R)R_L}$$

Substituting the component values,

$$H_A(s) = \frac{10000}{100 + 1000 + 10000 + s(68 \times 10^{-9})(1100)(10000)}$$

Therefore,

$$H_A(s) = \frac{10000}{11100 + 0.748s}$$

6.1.3 Voltage gain at the specified frequencies

The magnitude of the transfer function is

$$|H_A(j\omega)| = \frac{R_L}{\sqrt{(R_s + R + R_L)^2 + [\omega C(R_s + R)R_L]^2}}$$

At $1\ kHz$,

$$|H_A(j\omega_1)| = \frac{10000}{\sqrt{11100^2 + [6283.19(68 \times 10^{-9})(1100)(10000)]^2}}$$

$$= \frac{10000}{\sqrt{11100^2 + 4694.31^2}}$$

$$|H_A(j2\pi \times 1000)| \approx 0.830$$

At $10\ kHz$,

$$|H_A(j\omega_2)| = \frac{10000}{\sqrt{11100^2 + [62831.85(68 \times 10^{-9})(1100)(10000)]^2}}$$

$$= \frac{10000}{\sqrt{11100^2 + 46943.1^2}}$$

$$|\mathrm{H_A(j2\pi{\times}10000)}| \approx 0.207$$

Thus, Single-Stage meets both nominal gain requirements: at least 0.80 at 1 kHz and no more than 0.25 at 10 kHz.

6.1.4 ABCD parameters of Single-Stage

The transmission matrices of a series resistor and a shunt capacitor are

$$T_R = \begin{bmatrix} 1 & R \\ 0 & 1 \end{bmatrix}, \qquad T_C = \begin{bmatrix} 1 & 0 \\ sC & 1 \end{bmatrix}$$

Using the convention

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix}$$

where I_1enters port 1 and I_2 leaves port 2, the overall matrix is

$$T_A = T_R T_C$$

$$T_A = \begin{bmatrix} 1 & R \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ sC & 1 \end{bmatrix}$$

Multiplying the matrices,

$$T_A = \begin{bmatrix} 1 + sRC & R \\ sC & 1 \end{bmatrix}$$

Therefore,

$$A = 1 + sRC, \quad B = R, \quad C = sC, \quad D = 1$$

Substituting $R = 1000\,\Omega$ and $C = 68 \times 10^{-9}\,F$,

$$A = 1 + 68 \times 10^{-6}s$$
$$B = 1000\Omega$$
$$C = 68 \times 10^{-9}s\,S$$
$$D = 1$$

Here, C in the ABCD matrix denotes a transmission parameter, not the physical capacitor value.

6.1.5 Z-parameters of Single-Stage

For the Z-parameter representation, both port currents are defined as entering the two-port network:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

For the selected series-resistor and shunt-capacitor network,

$$Z_{11} = R + \frac{1}{sC}$$
$$Z_{12} = Z_{21} = Z_{22} = \frac{1}{sC}$$

Hence,

$$Z_A = \begin{bmatrix} R + \frac{1}{sC} & \frac{1}{sC} \\ \frac{1}{sC} & \frac{1}{sC} \end{bmatrix}$$

Substituting the component values,

$$Z_A = \begin{bmatrix} 1000 + \frac{1}{68 \times 10^{-9}s} & \frac{1}{68 \times 10^{-9}s} \\ \frac{1}{68 \times 10^{-9}s} & \frac{1}{68 \times 10^{-9}s} \end{bmatrix} \Omega$$

6.2. Dual-Stage RC Low-Pass Filter

6.2.1 Given parameters

$$R_s = 100\,\Omega$$
$$R_1 = R_2 = 330\,\Omega$$
$$C_1 = C_2 = 68 \times 10^{-9}\,F$$
$$R_L = 10000\,\Omega$$

6.2.2 Derivation of the ABCD parameters

The circuit consists of two cascaded RC sections. Its overall transmission matrix is

$$T_B = T_{R_1} T_{C_1} T_{R_2} T_{C_2} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

Using the series-resistor and shunt-capacitor matrices,

$$T_{R_i} = \begin{bmatrix} 1 & R_i \\ 0 & 1 \end{bmatrix}, \qquad T_{C_i} = \begin{bmatrix} 1 & 0 \\ sC_i & 1 \end{bmatrix}$$

Multiplication gives

$$A = 1 + s(R_1C_1 + R_1C_2 + R_2C_2) + s^2 R_1 R_2 C_1 C_2$$
$$B = R_1 + R_2 + sR_1R_2C_1$$
$$C = s(C_1 + C_2) + s^2 R_2 C_1 C_2$$
$$D = 1 + sR_2C_1$$

For the selected values,

$$R_1C_1 = R_1C_2 = R_2C_2 = 330(68 \times 10^{-9}) = 22.44 \times 10^{-6}$$

Thus,

$$A = 1 + 67.32 \times 10^{-6}s + 5.035536 \times 10^{-10}s^2$$

Similarly,

$$B = 330 + 330 + s(330)(330)(68 \times 10^{-9})$$
$$B = 660 + 7.4052 \times 10^{-3}s$$

For parameter C,

$$C = 136 \times 10^{-9}s + (330)(68 \times 10^{-9})^2 s^2$$
$$C = 136 \times 10^{-9}s + 1.52592 \times 10^{-12}s^2$$

Finally,

$$D = 1 + (330)(68 \times 10^{-9})s$$
$$D = 1 + 22.44 \times 10^{-6}s$$

Therefore, the ABCD matrix of Dual-Stage is

$$T_B = \begin{bmatrix} 1 + 67.32 \times 10^{-6}s + 5.035536 \times 10^{-10}s^2 & 660 + 7.4052 \times 10^{-3}s \\ 136 \times 10^{-9}s + 1.52592 \times 10^{-12}s^2 & 1 + 22.44 \times 10^{-6}s \end{bmatrix}$$

6.2.3 Transfer function of Dual-Stage

For the adopted ABCD convention, the output voltage is$V_o = V_2$, and the load current is $I_2 = V_o/R_L$.Therefore,

$$V_1 = \left(A + \frac{B}{R_L}\right) V_o$$

and

$$I_1 = \left(C + \frac{D}{R_L}\right) V_o$$

Since the source voltage includes the voltage drop across R_s,

$$V_s = V_1 + R_s I_1$$

Consequently,

$$H_B(s) = \frac{V_o}{V_s} = \frac{R_L}{AR_L + B + R_s(CR_L + D)}$$

Substituting the ABCD expressions and the source and load values gives

$$H_B(s) = \frac{10000}{10760 + 0.8188492s + 6.561456 \times 10^{-6}s^2}$$

This transfer function includes the source resistance and the nominal load resistance.

6.2.4 Voltage gain at the specified frequencies

For $s = j\omega$,

$$|H_B(j\omega)| = \frac{10000}{\sqrt{(10760 - 6.561456 \times 10^{-6}\omega^2)^2 + (0.8188492\omega)^2}}$$

At $1\,kHz$, $\omega_1 = 6283.19\,rad/s$:

$$|H_B(j\omega_1)| = \frac{10000}{\sqrt{(10760 - 6.561456 \times 10^{-6}(6283.19)^2)^2 + (0.8188492 \times 6283.19)^2}}$$

$$|H_B(j2\pi \times 1000)| \approx 0.855$$

At $10\,kHz$, $\omega_2 = 62831.85\,rad/s$:

$$|H_B(j\omega_2)| = \frac{10000}{\sqrt{(10760 - 6.561456 \times 10^{-6}(62831.85)^2)^2 + (0.8188492 \times 62831.85)^2}}$$

$$|H_B(j2\pi \times 10000)| \approx 0.186$$

The calculated gains meet both design requirements.

7. MATLAB Code

7.1 Complete MATLAB Program

```
clear; clc; close all;

%% 1. Given specifications and selected values
Rs = 100; RL = 10e3; Vs = 1;
f1 = 1e3; f2 = 10e3;
f = logspace(2,5,1000); s = 1j*2*pi*f;
gainMin1 = 0.80; gainMax10 = 0.25;

% Single-Stage: single-stage filter
RA = 1e3; CA = 68e-9;
% Dual-Stage: two-stage filter
R1B = 330; R2B = 330; C1B = 68e-9; C2B = 68e-9;

%% 2. Calculate transfer functions including Rs and RL
HA = singleStage(Rs,RA,CA,RL,s);
HB = doubleStage(Rs,R1B,C1B,R2B,C2B,RL,s);

HA1 = abs(singleStage(Rs,RA,CA,RL,1j*2*pi*f1));
HA10 = abs(singleStage(Rs,RA,CA,RL,1j*2*pi*f2));
HB1 = abs(doubleStage(Rs,R1B,C1B,R2B,C2B,RL,1j*2*pi*f1));
HB10 = abs(doubleStage(Rs,R1B,C1B,R2B,C2B,RL,1j*2*pi*f2));
```

```
fprintf('\nECE - 2108 Passive Two-Port Filter Project\n');
fprintf('Single-Stage: G(1 kHz)=%.6f, G(10 kHz)=%.6f\n',HA1,HA10);
fprintf('Dual-Stage: G(1 kHz)=%.6f, G(10 kHz)=%.6f\n',HB1,HB10);
fprintf('Single-Stage passes both nominal targets: %d\n',HA1>=gainMin1
&& HA10<=gainMax10);
fprintf('Dual-Stage passes both nominal targets: %d\n',HB1>=gainMin1 &&
HB10<=gainMax10);

%% 3. Combined magnitude and phase response figure
% One figure window with magnitude above and phase below.
figure('Color','w','Name','Combined Magnitude and Phase Response');

subplot(2,1,1);
semilogx(f,abs(HA),'LineWidth',1.8); hold on;
semilogx(f,abs(HB),'LineWidth',1.8);
yline(gainMin1,'--','1 kHz gain minimum = 0.80');
yline(gainMax10,'--','10 kHz gain maximum = 0.25');
xline(f1,':','1 kHz'); xline(f2,':','10 kHz');
grid on; xlim([100 1e5]);
xlabel('Frequency (Hz)'); ylabel('|V_o/V_s|');
title('Magnitude Response, R_L = 10 k\Omega');
legend('Single-Stage:       single       stage','Dual-Stage:       two
stages','Location','southwest');

subplot(2,1,2);
semilogx(f,angle(HA)*180/pi,'LineWidth',1.8); hold on;
semilogx(f,angle(HB)*180/pi,'LineWidth',1.8);
xline(f1,':','1 kHz'); xline(f2,':','10 kHz');
grid on; xlim([100 1e5]);
xlabel('Frequency (Hz)'); ylabel('Phase (degrees)');
title('Phase Response, R_L = 10 k\Omega');
legend('Single-Stage:       single       stage','Dual-Stage:       two
stages','Location','southwest');

%% 8. Load variation: equation-based predictions and supplied Multisim
values
RLvals = [5e3 10e3 20e3];
loadA1 = zeros(3,1); loadA10 = zeros(3,1);
loadB1 = zeros(3,1); loadB10 = zeros(3,1);
for k = 1:3
    loadA1(k) = abs(singleStage(Rs,RA,CA,RLvals(k),1j*2*pi*f1));
    loadA10(k) = abs(singleStage(Rs,RA,CA,RLvals(k),1j*2*pi*f2));
    loadB1(k)                                                    =
abs(doubleStage(Rs,R1B,C1B,R2B,C2B,RLvals(k),1j*2*pi*f1));
```

```
    loadB10(k)                                                          =
abs(doubleStage(Rs,R1B,C1B,R2B,C2B,RLvals(k),1j*2*pi*f2));
end
figure('Color','w','Name','Load Variation');
subplot(1,2,1);
plot(RLvals/1e3,loadA1,'o-','LineWidth',1.6); hold on;
plot(RLvals/1e3,loadB1,'s-','LineWidth',1.6); yline(gainMin1,'--');
grid on; xlabel('R_L (k\Omega)'); ylabel('Gain at 1 kHz');
title('Useful-Signal  Gain  vs  Load');  legend('Single  stage','Two
stages','Limit','Location','best');
subplot(1,2,2);
plot(RLvals/1e3,loadA10,'o-','LineWidth',1.6); hold on;
plot(RLvals/1e3,loadB10,'s-','LineWidth',1.6); yline(gainMax10,'--');
grid on; xlabel('R_L (k\Omega)'); ylabel('Gain at 10 kHz');
title('Interference  Gain  vs  Load');  legend('Single  stage','Two
stages','Limit','Location','best');

%% 9. ±5% sensitivity analysis, nominal load
% Each variation is applied to one component at a time.
variation = ["R -5%";"R +5%";"C -5%";"C +5%"];
RAvar = [0.95*RA;1.05*RA;RA;RA];
CAvar = [CA;CA;0.95*CA;1.05*CA];
R1var = [0.95*R1B;1.05*R1B;R1B;R1B];
C1var = [C1B;C1B;0.95*C1B;1.05*C1B];
SA1=zeros(4,1); SA10=zeros(4,1); SB1=zeros(4,1); SB10=zeros(4,1);
for k=1:4
    SA1(k)=abs(singleStage(Rs,RAvar(k),CAvar(k),RL,1j*2*pi*f1));
    SA10(k)=abs(singleStage(Rs,RAvar(k),CAvar(k),RL,1j*2*pi*f2));

SB1(k)=abs(doubleStage(Rs,R1var(k),C1var(k),R2B,C2B,RL,1j*2*pi*f1));

SB10(k)=abs(doubleStage(Rs,R1var(k),C1var(k),R2B,C2B,RL,1j*2*pi*f2));
end
figure('Color','w','Name','Sensitivity Analysis');
subplot(1,2,1); bar([SA1 SB1]); grid on;
set(gca,'XTickLabel',variation);    ylabel('Gain    at    1    kHz');
title('Sensitivity at 1 kHz');
legend('Single stage','Two stages','Location','best');
subplot(1,2,2); bar([SA10 SB10]); grid on;
set(gca,'XTickLabel',variation);    ylabel('Gain    at    10    kHz');
title('Sensitivity at 10 kHz');
legend('Single stage','Two stages','Location','best');

%% 10. Independent hand verification: Single-Stage
handGain = @(freq) RL ./ sqrt((Rs+RA+RL).^2 + ...
```

```
    (2*pi*freq*CA*(Rs+RA)*RL).^2);
hand1 = handGain(f1); hand10 = handGain(f2);
multisim1 = 0.830; multisim10 = 0.207;
fprintf('Independent hand verification, Single-Stage:\n');
fprintf('Gain     at     1     kHz:     %.6f     (recorded     Multisim:
%.3f)\n',hand1,multisim1);
fprintf('Gain     at     10     kHz:     %.6f     (recorded     Multisim:
%.3f)\n',hand10,multisim10);
fprintf('For Vs = %.2f V, calculated output amplitudes: %.6f V at 1 kHz
and %.6f V at 10 kHz.\n', ...
    Vs,Vs*hand1,Vs*hand10);

%% 11. Required design revision
% Example revision: increase Single-Stage capacitor by 5%, from 68 nF
to 71.4 nF.
% This is a documented trial revision, not claimed to be globally
optimal.
CAnew = 1.05*CA;
HArev = singleStage(Rs,RA,CAnew,RL,s);
rev1 = abs(singleStage(Rs,RA,CAnew,RL,1j*2*pi*f1));
rev10 = abs(singleStage(Rs,RA,CAnew,RL,1j*2*pi*f2));
figure('Color','w','Name','Design Revision');
semilogx(f,abs(HA),'LineWidth',1.8); hold on;
semilogx(f,abs(HArev),'LineWidth',1.8);
yline(gainMin1,'--','Minimum gain 0.80'); yline(gainMax10,'--','Maximum
gain 0.25');
xline(f1,':','1 kHz'); xline(f2,':','10 kHz');
grid on; xlim([100 1e5]); xlabel('Frequency (Hz)'); ylabel('|V_o/V_s|');
title('Design Revision: C increased by 5%');
legend('Original     C     =     68     nF','Revised     C     =     71.4
nF','Location','southwest');

%% 12. ABCD parameters and Z parameters
% ABCD convention: [V1; I1] = [A B; C D]*[V2; I2].
% I1 enters port 1; I2 leaves port 2. Rs and RL are external to the two-
port.
fprintf('\nABCD parameters, Single-Stage:\n');
fprintf('A = 1 + s*R*C = 1 + s*(%.9g)\n',RA*CA);
fprintf('B = R = %.9g ohm\n',RA);
fprintf('C = s*C = s*(%.9g) S\n',CA);
fprintf('D = 1\n');

fprintf('\nABCD parameters, Dual-Stage:\n');
fprintf('A = 1 + s*(R1*C1 + R1*C2 + R2*C2) + s^2*R1*R2*C1*C2\n');
```

```
fprintf('                =        1        +        s*(%.9g)        +
s^2*(%.9g)\n',R1B*C1B+R1B*C2B+R2B*C2B,R1B*R2B*C1B*C2B);
fprintf('B   =   R1   +   R2   +   s*R1*R2*C1   =   %.9g   +   s*(%.9g)
ohm\n',R1B+R2B,R1B*R2B*C1B);
fprintf('C   =   s*(C1+C2)   +   s^2*R2*C1*C2   =   s*(%.9g)   +   s^2*(%.9g)
S\n',C1B+C2B,R2B*C1B*C2B);
fprintf('D = 1 + s*R2*C1 = 1 + s*(%.9g)\n',R2B*C1B);

% Z parameters for Single-Stage, with both port currents entering the
network:
% Z11 = R + 1/(sC); Z12 = Z21 = Z22 = 1/(sC).
fprintf('\nZ   parameters   for   Single-Stage   (both   port   currents
entering):\n');
fprintf('Z11(s) = %.9g + 1/(s*%.9g) ohm\n',RA,CA);
fprintf('Z12(s) = Z21(s) = Z22(s) = 1/(s*%.9g) ohm\n',CA);

fprintf('Figures generated: magnitude/phase response, load variation,
sensitivity analysis, and design revision.\n');

%% Local functions (MATLAB R2016b or newer)
function H = singleStage(Rs,R,C,RL,s)
    H = RL ./ (Rs + R + RL + s.*C.*(Rs+R).*RL);
end

function H = doubleStage(Rs,R1,C1,R2,C2,RL,s)
    % Exact nodal solution. Rs and R1 are in the source-to-node-1 path.
    % Node 1 is connected to C1 and R2; output is loaded by RL || C2.
    Gs = 1/(Rs+R1);
    G2 = 1/R2;
    Yout = 1/RL + s.*C2;
    denominator = (Gs+s.*C1+G2).*(G2+Yout) - G2^2;
    H = (Gs*G2)./denominator;
end
```

7.2 Program Outputs and Simulation Graphs

```
Command Window

ECE - 2108 Passive Two-Port Filter Project
Design A: G(1 kHz)=0.829602, G(10 kHz)=0.207077
Design B: G(1 kHz)=0.855166, G(10 kHz)=0.186455
Design A passes both nominal targets: 1
Design B passes both nominal targets: 1
Independent hand verification, Design A:
Gain at 1 kHz: 0.829602 (recorded Multisim: 0.830)
Gain at 10 kHz: 0.207077 (recorded Multisim: 0.207)
For Vs = 1.00 V, calculated output amplitudes: 0.829602 V at 1 kHz and 0.207077 V at 10 kHz.

ABCD parameters, Design A:
A = 1 + s*R*C = 1 + s*(6.8e-05)
B = R = 1000 ohm
C = s*C = s*(6.8e-08) S
D = 1

ABCD parameters, Design B:
A = 1 + s*(R1*C1 + R1*C2 + R2*C2) + s^2*R1*R2*C1*C2
  = 1 + s*(6.732e-05) + s^2*(5.035536e-10)
B = R1 + R2 + s*R1*R2*C1 = 660 + s*(0.0074052) ohm
C = s*(C1+C2) + s^2*R2*C1*C2 = s*(1.36e-07) + s^2*(1.52592e-12) S
D = 1 + s*R2*C1 = 1 + s*(2.244e-05)

Z parameters for Design A (both port currents entering):
Z11(s) = 1000 + 1/(s*6.8e-08) ohm
Z12(s) = Z21(s) = Z22(s) = 1/(s*6.8e-08) ohm
Figures generated: magnitude/phase response, load variation, sensitivity analysis, and design revision.
```

Fig-03: Program Output for ABCD and Z Parameters

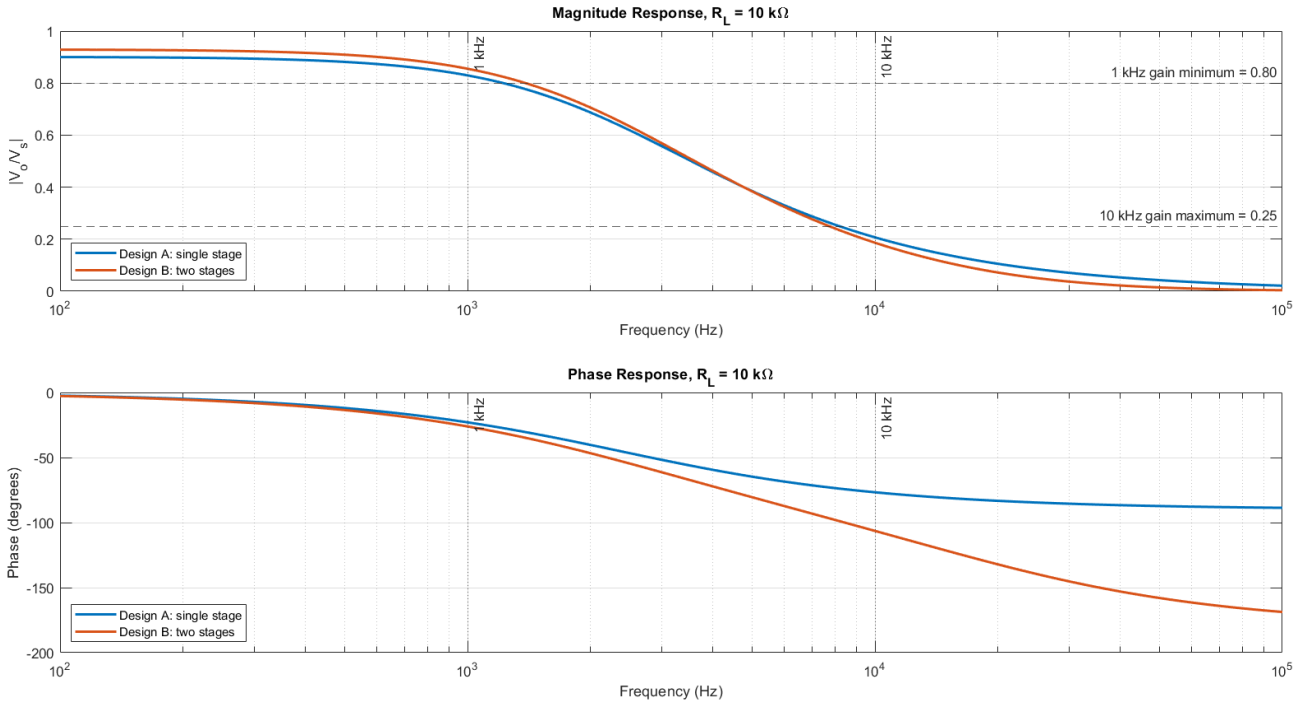
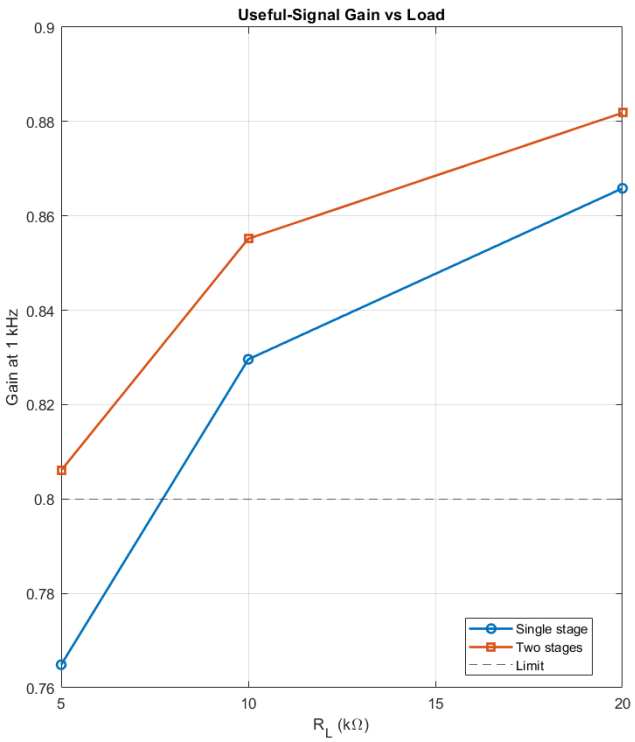


Fig-04: Combined Magnitude and Phase Response



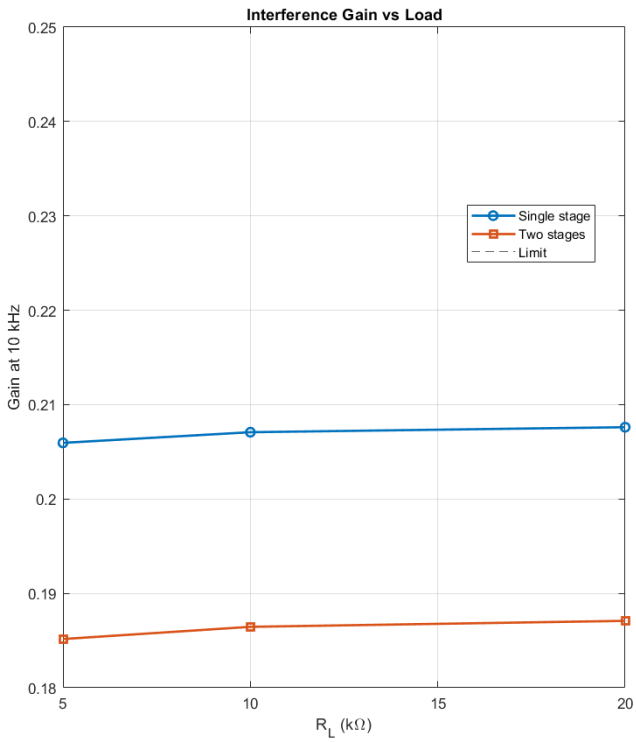
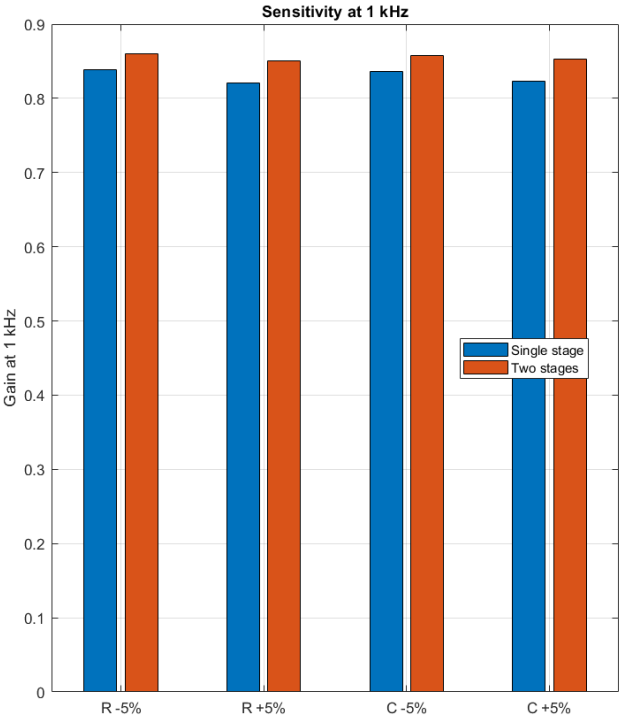


Fig-05: Load Variation in both RC Filters



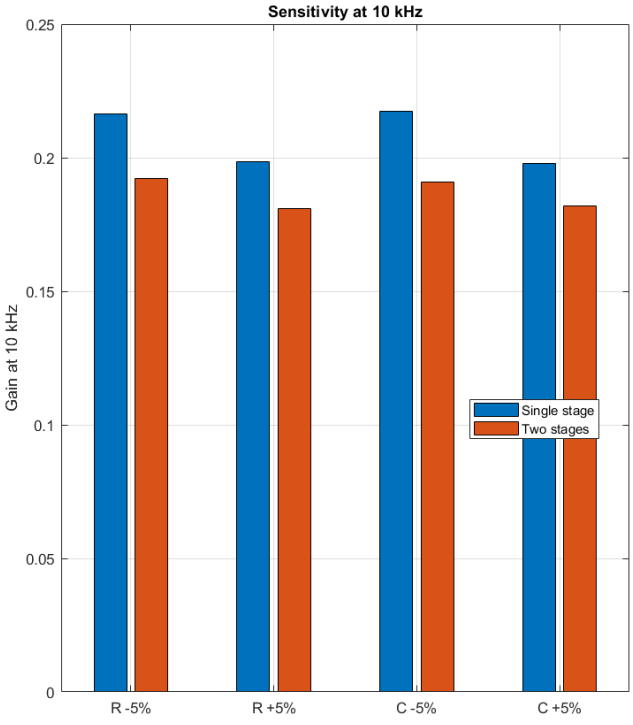


Fig-06: 5% Sensitivity Analysis in both RC Filters

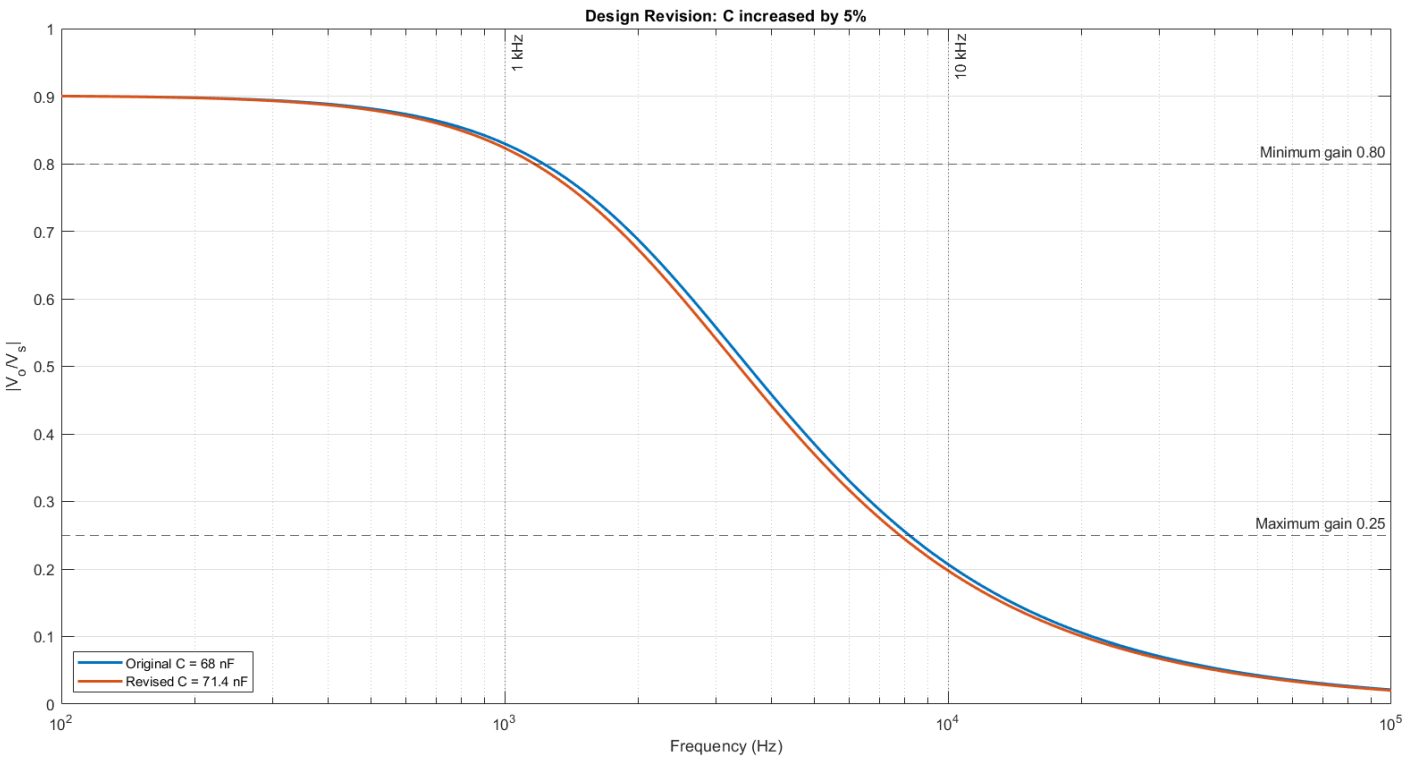


Fig-07: Design Revision for both RC Filters

8. Result Analysis

8.1 Selection of Component Values

Several resistor and capacitor combinations were evaluated to identify suitable component values for the two proposed passive low-pass filters. The selection criteria were a voltage gain magnitude of at least 0.80 at 1 kHz and no more than 0.25 at 10 kHz, with a source resistance of $100\ \Omega$ and nominal load resistance of $10\ k\Omega$.

For the single-stage RC filter, ten component combinations were evaluated. The recorded results indicate that only two combinations satisfy both gain requirements: $R = 1\ k\Omega$, $C = 68\ nF$, and $R = 1.5\ k\Omega$, $C = 47\ nF$. The first combination was selected because it provides a useful-signal gain of 0.830 and an interference-frequency gain of 0.207, satisfying both specifications with a useful-signal gain margin of 0.030.

For the two-stage RC filter, the selected component values were $R_1 = R_2 = 330\ \Omega$ and $C_1 = C_2 = 68\ nF$. The recorded candidate results for this combination are 0.855 at 1 kHz and 0.186 at 10 kHz. These values satisfy both design requirements. The candidate table should be identified as calculated or simulated according to the actual source of those values.

The selected component values and nominal results are summarized below.

	Single-Stage	Dual-Stage
Resistor values	$1\ k\Omega$	$330\ \Omega$ each
Capacitor values	$68\ nF$	$68\ nF$ each
Gain at 1 kHz	0.830	0.855
Gain at 10 kHz	0.207	0.186
Component count	2	4
Nominal requirements	Satisfied	Satisfied

Single-Stage has the advantage of a lower component count and a simpler circuit. Dual-Stage provides a higher recorded useful-signal gain and stronger recorded interference rejection. Therefore, Dual-Stage is the more attractive option when interference suppression is prioritized, whereas Single-Stage is preferable when circuit simplicity is the primary consideration.

8.2 Comparison of Frequency Responses

The MATLAB magnitude and phase plots provide a frequency-domain comparison of the two filter configurations over the required range of 100 Hz to 100 kHz. Both filters exhibit low-pass behavior: the output gain decreases as the input frequency increases, while the phase response becomes increasingly negative.

At the useful frequency of 1 kHz, the recorded gain of Single-Stage is 0.830, compared with 0.855 for Dual-Stage. The corresponding output amplitudes for a 1V sinusoidal source are approximately 0.830 V and 0.855 V, respectively.

At the interference frequency of 10 kHz, the recorded gain decreases to 0.207 for Single-Stage and 0.186 for Dual-Stage. For a 1 V input, these correspond to output amplitudes of approximately 0.207 V and 0.186 V. Thus, the recorded results indicate that Dual-Stage attenuates the unwanted component more effectively at 10 kHz.

The two-stage configuration uses the maximum permitted number of filter components, while the single-stage design uses only two. The improvement in interference rejection must therefore be considered alongside the increased component count and circuit complexity.

The MATLAB plots should be used to demonstrate the complete response rather than relying only on the two specified frequencies. The nominal gains quoted above are the previously recorded values; the calculated transfer-function response and MATLAB plot should be checked for consistency before final submission.

8.3 Effect of Load Resistance

The project requires the performance of both filters to be investigated at load resistances of $5\ k\Omega$, $10\ k\Omega$, and $20\ k\Omega$. The recorded results are shown below.

Load resistance	Single-Stage, **1 kHz**	Single-Stage, **10 kHz**	Dual-Stage, **1 kHz**	Dual-Stage, **10 kHz**
$5\ k\Omega$	0.765	0.206	0.806	0.185
$10\ k\Omega$	0.830	0.207	0.855	0.186
$20\ k\Omega$	0.865	0.207	0.885	0.200

The results show that increasing the load resistance increases the recorded gain at 1 kHz for both designs. This occurs because a higher load resistance reduces the loading effect on the output. Conversely, reducing the load resistance increases the loading effect and decreases the useful-signal gain.

At $5\ k\Omega$, Single-Stage produces a gain of 0.765 at 1 kHz, which falls below the required minimum of 0.80. Although its recorded gain at 10 kHz remains within the interference limit, the design does not satisfy both requirements at this load. Dual-Stage produces a gain of 0.806 at 1 kHz and 0.185 at 10 kHz, meeting both targets under the same recorded conditions.

At the nominal $10\ k\Omega$ load, both designs satisfy the specified requirements. At $20\ k\Omega$, both also remain within the recorded gain limits. Therefore, Dual-Stage demonstrates better compliance across the three tested loads, while Single-Stage exhibits a limitation under heavier loading.

8.4 Component Tolerance and Design Revision

The effect of component tolerances was investigated by changing one resistor or capacitor by $\pm 5\%$while keeping the remaining components at their nominal values. This analysis evaluates whether normal component variations could cause either filter to violate the gain specifications.

Variation	**Single-Stage: 1 kHz**	**Single-Stage: 10 kHz**	**Dual-Stage: 1 kHz**	**Dual-Stage: 10 kHz**
Resistor −5%	0.838	0.216	0.863	0.196
Resistor +5%	0.820	0.198	0.847	0.177
Capacitor −5%	0.835	0.217	0.854	0.188
Capacitor +5%	0.823	0.198	0.840	0.167

All entries in this recorded table satisfy the two nominal gain limits. The results suggest that both designs can tolerate the investigated component changes under the nominal-load conditions. However, the effect of resistor and capacitor variation differs between the designs, and the magnitude of the change should be assessed from the corresponding frequency-response results.

A design revision was also considered by increasing the capacitance of Single-Stage by 5%, from $68\,nF$to $71.4\,nF$, while keeping the resistor and load unchanged. Using the Single-Stage transfer function, the revised gains are approximately

$$|H_A(j2\pi \times 1000)| \approx 0.823$$

$$|H_A(j2\pi \times 10000)| \approx 0.198$$

Compared with the original calculated values of approximately 0.830 and 0.207, respectively, the revised capacitor slightly reduces the useful-signal gain while improving interference attenuation. Both gain requirements remain satisfied at the nominal load. This revision demonstrates the trade-off between preserving the useful signal and suppressing higher-frequency interference.

If the interference frequency were reduced from 10 kHz to 5 kHz, the gain at 5 kHz would need to be evaluated using the MATLAB response. Because a low-pass filter generally attenuates a 5 kHz component less than a 10 kHz component, the existing designs might provide insufficient rejection at the lower interference frequency. The component values would then need to be reconsidered against the revised specification.

Overall, the recorded results favor Dual-Stage when load robustness and interference suppression are the main priorities. Single-Stage remains a simpler alternative, but its lower useful-signal gain at a $5\,kOmega$load is a limitation. The final choice should be supported by the verified MATLAB plots and consistent simulation or circuit-calculation results.

9. Discussion

The project investigated two passive RC low-pass filter configurations to preserve a useful signal at 1 kHz while suppressing unwanted interference at 10 kHz. Both designs satisfied the specified nominal gain requirements according to the recorded results. Single-Stage offered a simpler implementation with only two filter components, whereas Dual-Stage used four components to achieve greater recorded interference attenuation.

The load-variation analysis demonstrated that the load resistance affects the useful-signal gain of both filters. In particular, Single-Stage failed to meet the minimum gain requirement at a $5\,k\Omega$ load, while Dual-Stage satisfied both requirements under the recorded conditions. This indicates that Dual-Stage is a more suitable choice when performance across different load resistances is important.

The component-tolerance analysis showed that the recorded results remained within the specified gain limits for the investigated $\pm 5\%$variations. The design revision further illustrated the trade-off between useful-signal transmission and interference rejection: increasing the capacitance of Single-Stage improved attenuation at 10 kHz but slightly reduced the gain at 1 kHz.

Overall, Dual-Stage appears preferable when interference suppression and load robustness are prioritized, while Single-Stage remains attractive because of its simplicity and lower component count. The final selection should be supported by consistent MATLAB and circuit-simulation results, particularly when comparing the predicted responses with the recorded measurements.

10. Conclusion

The project successfully investigated the Single-Stagend MATLAB simulation of two passive RC low-pass filters for passing a useful 1 kHz signal while attenuating unwanted 10 kHz interference. Both designs satisfied the specified gain requirements under nominal conditions. Single-Stage offered a simpler circuit with fewer components, whereas Dual-Stage provided better interference rejection and more consistent performance under different load resistances.

The load variation analysis showed that Single-Stage failed to maintain the required gain at 1 kHz when the load resistance decreased to 5 kΩ, while Dual-Stage continued to satisfy the requirements. Increasing Single-Stage's capacitance from 68 nF to 71.4 nF improved interference attenuation but slightly reduced the useful-signal gain. If the interference frequency changes from 10 kHz to 5 kHz, the filter response must be re-evaluated and the component values adjusted if necessary. Overall, Dual-Stage is the preferred option when filtering performance and load stability are prioritized, while Single-Stage remains suitable when circuit simplicity and lower component count are more important.

11. References

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